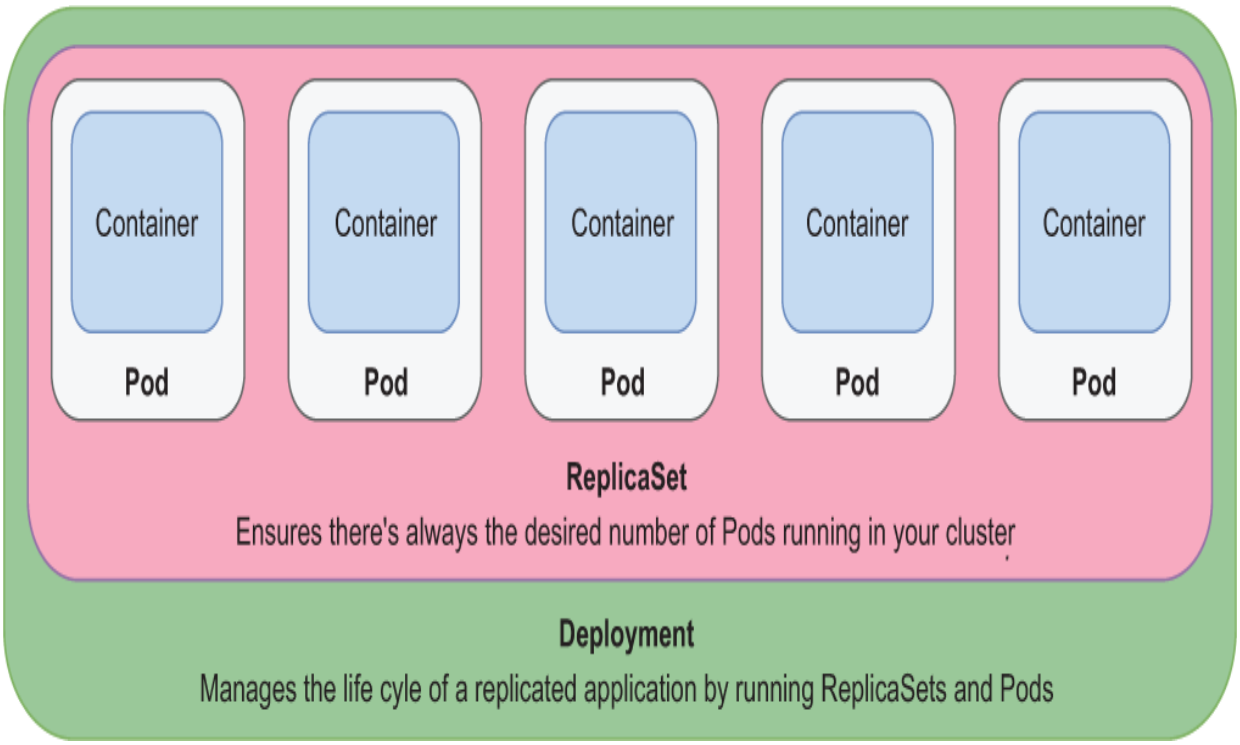


Figure 7.4 A Deployment manages replicated applications with ReplicaSets and Pods in your cluster. A ReplicaSet ensures the desired number of Pods is always up and running. A Pod runs a containerized application.

Deployments provide a convenient abstraction for us to declare what we want to achieve (the *desired state*), and we can let Kubernetes make it happen. You don't need to worry about how to achieve a specific result. Unlike imperative tools such as Ansible or Puppet, you can just tell Kubernetes what you want, and the orchestrator will figure out how to achieve the desired result and keep it consistent. That's what we call *declarative configuration*.

Kubernetes uses controllers that watch the system and compare the desired state with the actual state. When there is any difference between the two, it acts to make them match again. Deployments and ReplicaSets are controller objects, handling rollout, replication, and self-healing. For example, suppose